Spectral Signatures of Inductive Bias: Analyzing Attention Mechanisms from Synthetic to Realistic Data

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Abstract

This work investigates the impact of structural priors on the learning dynamics of Transformer attention mechanisms. Through a comparative analysis of Tied, Untied, and Unfactorized parameterizations, we explore how architectural choices shape generalization in two complementary settings: a controlled high-dimensional synthetic framework and a structured discrete histogram task. Our results reveal a clear hierarchy of inductive biases , demonstrating that the Positive Semi-Definite (PSD) geometry inherent to tied attention is critical for learning semantic similarity, significantly improving data efficiency compared to both unstructured full-rank and over-parameterized asymmetric alternatives, even when symmetry is explicitly enforced. Finally, we link these functional differences to spectral signatures, identifying signal-to-noise phase transitions in the eigenvalue distributions.

1 Introduction

The Transformer architecture [1] has fundamentally reshaped machine learning, establishing state-of-the-art performance across natural language processing [2, 3], computer vision [4], and multimodal reasoning. At its core lies the self-attention mechanism, a flexible compatibility kernel that enables the modeling of long-range dependencies between tokens. While the success of Transformers is undeniable, the theoretical principles governing their specific structural choices (such as the low-rank factorization into Queries (Q) and Keys (K)) remain poorly understood. It remains an open question whether this factorization serves merely as a computational bottleneck or as a fundamental inductive bias that guides the optimization process. Recent works suggest that this factorization is not merely an architectural constraint but plays a crucial role in adaptation and generalization [5, 6].

A growing body of empirical evidence suggests that the weight matrices in trained Transformers develop pronounced spectral structures, characterized by low-rank components and heavy-tailed eigenvalue distributions [7, 8, 9]. These spectral signatures reflect implicit regularization mechanisms [10, 11, 12] that correlate strongly with generalization capabilities [13]. In this framework, spectral signatures are more than mere empirical correlations; they act as order parameters signaling a fundamental phase transition between a regime of brute-force noise memorization and one of emergent structural discovery. Recent theoretical works have begun to formalize these observations using simplified frameworks in high-dimensional settings. Notably, [14] provided Bayes-optimal characterizations of tied attention, while [15] highlighted the inductive biases inherent in tied versus untied parameterizations. Similarly, [16] revealed phase transitions between positional and semantic attention patterns using solvable models.

Despite these advances, a gap remains between the idealized Bayes-optimal limits and the practical dynamics of Empirical Risk Minimization (ERM) on finite samples [17, 18]. Specifically, what is the impact of the structural prior on learning efficiency? Is the factorized structure ($QK^\top$) merely a computational convenience, or does it impose a necessary inductive bias for learning semantic similarities, as typically observed in linear networks [19]?

To address these questions, we analyze the learning dynamics of Tied (Symmetric PSD), Untied, and Unfactorized attention matrices across two complementary regimes. We begin with the synthetic high-dimensional limit, using a controlled Teacher-Student framework with Gaussian inputs to rigorously derive the generalization error as a function of sample complexity $\alpha = n/d^2$. This setting connects directly to the fundamental limits of matrix sensing [20] and random matrix theory [21, 22]. We then contrast these theoretical insights with a structured discrete task (histogram task) inspired by [23], which requires exact counting capabilities for discrete sequences and introduces the practical complexities of learnable embeddings and non-linear MLP readout heads.

All code and experiments are available at: https://github.com/victor-pcll/Exploring-transformers-empirically.

2 Synthetic Dataset Analysis

In this section, we investigate the learning dynamics of different attention mechanisms. We specifically focus on how structural priors influence generalization performance in the high-dimensional regime.

2.1 Setting and Scaling Limit

To enable analytical tractability, we place ourselves in the high-dimensional limit, where the token embedding dimension $d \to \infty$. We assume that the number of training samples n scales quadratically with d, and the rank of the target function weights $r^\star$ grows proportionally to d. Formally, we study the joint limit $d, n, r^\star \to \infty$ with fixed ratios:

$$\rho^\star = \frac{r^\star}{d} = \mathcal{O}(1), \qquad \alpha = \frac{n}{d^2} = \mathcal{O}(1) \tag{1}$$

The quadratic scaling of the sample size (α) is primarily driven by complexity constraints, as the number of degrees of freedom required to fully specify the target interaction matrix $S^\star \in \mathbb{R}^{d\times d}$ grows quadratically with d. Additionally, this scaling is necessary to capture the phase transition behavior, ensuring that the learning performance evolves from trivial to near-perfect within a finite range of α.

2.2 Synthetic Data Generation

We consider a supervised learning setting where the dataset $\mathcal{D} = \{(x^\mu, y^{\star\mu})\}_{\mu=1}^n$ is generated synthetically to ensure analytical tractability.

Input Distribution. The input sequences $x^\mu \in \mathbb{R}^{T\times d}$ consist of T tokens, where each token x_a^μ (for $1 \le a \le T$) is drawn independently from a high-dimensional standard Gaussian distribution:

$$x_a^\mu \sim \mathcal{N}(0, \mathbb{I}_d) \tag{2}$$

We derive the explicit moments and variance of quadratic forms under this Gaussian assumption in Appendix A, establishing the theoretical stability of the attention mechanism. While we assume Gaussian inputs for theoretical convenience, we note that the high-dimensional regime ($d \to \infty$) exhibits a universality property. As discussed in recent works [24, 25], the spectral properties of the learned matrices tend to depend only on the first two moments of the data distribution.

Target Generation (Teacher Model). The target labels $y^{\star\mu}$ are generated by a ground-truth "Teacher" attention mechanism, parameterized by a matrix $S^\star$. The attention scores h_{ab}^μ are computed as a centered quadratic form:

$$h_{ab}^\mu = \frac{(x_a^\mu)^\top S^\star x_b^\mu - \delta_{ab}\operatorname{Tr}(S^\star)}{\sqrt{d}} \tag{3}$$

The subtraction of the trace term $\delta_{ab}\operatorname{Tr}(S^\star)$ ensures that the diagonal entries are centered around zero. Under the Gaussian assumption, this term concentrates to the expectation of the diagonal (see derivations in Appendix A). The final targets $y^{\star\mu}$ are obtained by corrupting these scores with additive Gaussian noise before applying the softmax nonlinearity σ_β:

$$y^{\star\mu} = \sigma_\beta(h^\mu + \sqrt{\Delta}\,\xi^\mu) \tag{4}$$

where $\xi_{ab}^\mu \sim \mathcal{N}(0,1)$ represents the random noise fluctuations, β denotes the inverse temperature and $\Delta \ge 0$ controls the noise variance. Before analyzing the impact of priors, we validated our Teacher-Student generation pipeline by reproducing established Bayes-optimal results for the Tied architecture in Appendix B. To study the impact of the structural prior, we define three distinct structures for the Teacher matrix $S^\star$:

$$S^\star = \begin{cases} \frac{1}{\sqrt{r^\star d}} W^\star {W^\star}^\top & \text{Tied (Factorized)} \\ \frac{1}{\sqrt{r^\star d}} W_Q^\star {W_K^\star}^\top & \text{Untied} \\ S_{\text{unfac}}^\star & \text{Unfactorized} \end{cases} \tag{5}$$

where the factor matrices $W^\star, W_K^\star, W_Q^\star \in \mathbb{R}^{d\times r^\star}$ consist of i.i.d. standard Gaussian entries, with the rank fixed at $r^\star = \rho^\star d$. The entries of the full-rank matrix $S_{\text{unfac}}^\star$ are drawn independently from $\mathcal{N}(0, 1/d)$.

2.3 Student Architecture

Student Model. The student model seeks to approximate the teacher's mapping using a learnable attention mechanism parameterized by $S \in \mathbb{R}^{d\times d}$. To ensure a fair comparison, particularly for the Tied configuration, we apply the same zero-diagonal constraint as the teacher. The estimated outputs $\hat{y}^\mu$ are obtained via the softmax nonlinearity.

Parameterization. We investigate three distinct parameterizations mirroring the teacher structures:

$$S = \begin{cases} \frac{1}{\sqrt{rd}} W W^\top & \text{Tied (Factorized)} \\ \frac{1}{\sqrt{rd}} W_Q W_K^\top & \text{Untied} \\ S_{\text{unfac}} & \text{Unfactorized} \end{cases} \tag{6}$$

where the capacity is controlled by the width ratio $\rho = r/d$. Throughout our experiments, we maintain a matched architectural setting, where the student parameterization corresponds to the generative class of the teacher (e.g., an Unfactorized student learning from an Unfactorized teacher). This experimental design isolates the optimization dynamics unique to each architecture, allowing us to identify which structural prior promotes superior data efficiency under realizable conditions. Furthermore, we specifically examine the over-parameterized regime for the Untied model, where the student's width ratio ρ exceeds the teacher's rank ratio $\rho^\star$ (i.e., $\rho > \rho^\star$), to evaluate how implicit regularization influences generalization despite the increased capacity.

2.4 Algorithm

We frame the learning problem as the minimization of the regularized Sum of Squared Errors (SSE). Let Θ denote the set of learnable parameters. The estimator $\hat{\Theta}$ is obtained by minimizing:

$$\hat{\Theta} = \arg\min_{\Theta} \left[\sum_{\mu=1}^{n} \| y^{\star\mu} - \hat{y}^{\mu}(x^{\mu}, \Theta) \|_F^2 + \lambda r(\Theta) \right] \tag{7}$$

where $r(\Theta)$ is the Frobenius norm penalty adapted to the architecture (Tied, Untied, or Unfactorized) and λ is the regularization strength (or weight decay coefficient).

Initialization and Training. To ensure a fair comparison across architectures, we align the initial variance of the interaction matrix entries to $1/d$. For factorized models, the scaling factor $1/\sqrt{rd}$ combined with standard normal weights yields an effective variance of $\mathrm{Var}(S_{ij}) \approx \frac{1}{rd} \cdot r = 1/d$, matching the explicit initialization of the unfactorized model:

$$W_{ij} \sim \mathcal{N}(0, 1), \quad S_{\text{unfac},ij} \sim \mathcal{N}(0, 1/d) \tag{8}$$

This standardization ensures that all models start with equivalent signal amplitudes and gradient scales. Training is performed using Adam Optimizer [26] with a learning rate η, continuing until the absolute difference in total loss between consecutive epochs falls below a specified tolerance threshold (tol).

Evaluation. Generalization performance is quantified by the error ε_{gen}, calculated as the Mean Squared Error (MSE) over an independent test set of n_{test} samples:

$$\varepsilon_{\text{gen}} = \frac{1}{n_{\text{test}}} \sum_{\mu=1}^{n_{\text{test}}} \| y^{\star\mu} - \hat{y}^{\mu}(\hat{\Theta}) \|_F^2 \tag{9}$$

2.5 Results

Figure 1 presents the generalization error as a function of sample complexity α for the Unfactorized and Untied architectures under matched architectural setting. This comparison highlights the divergent learning dynamics between unstructured and structured low-rank interactions. Notably, while both architectures possess equivalent full-rank capacity ($r = d$), we contrast them to evaluate how their distinct optimization trajectories influence generalization performance.

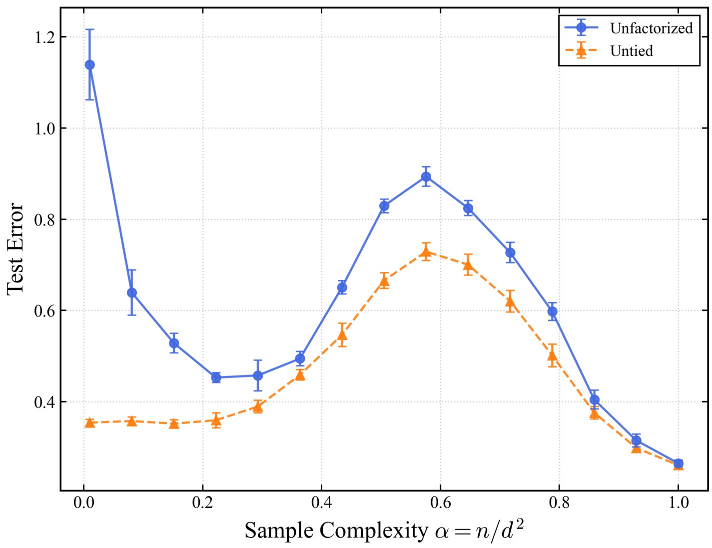


Figure 1: **Comparison of Generalization Error in Matched Settings.** Test error is plotted as a function of sample complexity $\alpha = n/d^2$. (1) **Unfactorized:** Student Unfactorized learning from Teacher Unfactorized. (2) **Untied:** Student Untied ($\rho = 1.0$) learning from Teacher Untied ($\rho^\star = 0.5$). **Observation:** Both asymmetric architectures exhibit a synchronized generalization peak around $\alpha \in [0.5, 0.6]$. However, the Unfactorized model suffers from a higher error floor, reflecting the higher entropy of the Unfactorized learning task compared to the Untied task. **Settings:** $T = 2$ tokens, $d = 100$, optimized via Adam ($\eta = 0.1$, tol $= 10^{-6}$). Results averaged over 16 experiments with $\lambda = 10^{-4}$ noise $\Delta = 0.5$ and a fixed number of test samples $n_{\text{test}} = 2\,000$.

Synchronized Peak. We observe a striking alignment in the critical regime across both asymmetric models. Both the Unfactorized and Untied architectures exhibit a generalization cusp within the same interval of $\alpha \in [0.5, 0.6]$. Crucially, this interpolation peak remains stable at nearly the same sample complexity for various width ratios $\rho \in [0.5, 1.5]$, as detailed in Appendix D. This stability suggests that for asymmetric parameterizations (whether full-rank or factorized) the transition between the under-parameterized and over-parameterized regimes is governed by a consistent complexity threshold. This behavior contrasts sharply with the Tied architecture, which reaches its peak much earlier at $\alpha \approx 0.2$ (Appendix C). Such a discrepancy indicates that the Positive Semi-Definite (PSD) constraint, fundamental to the Tied model, effectively reduces model capacity and shifts the interpolation threshold toward a lower sample complexity compared to unconstrained variants.

Complexity Gap. Despite the alignment of their peaks, a significant performance gap persists in the $\alpha < 1$ regime, where the Unfactorized model consistently yields higher error than its Untied counterpart. This difference highlights the intrinsic difficulty of their respective tasks: while the Unfactorized student must learn a target with d^2 independent degrees of freedom, the Untied student targets a structured low-rank interaction with an intrinsic dimensionality of only $2\rho^\star d^2$. Furthermore, as demonstrated in Appendix D, the Untied architecture maintains this performance advantage across all studied width ratios ($\rho > \rho^\star$), consistently outperforming the Unfactorized baseline throughout the $\alpha < 1$ regime.

3 Structured Discrete Data Analysis: The Histogram Task

While the Gaussian setting allows for precise theoretical predictions, real-world attention mechanisms operate on discrete tokens. In this section, we investigate the learning behavior of our models on a "Histogram" task, which requires the network to perform exact counting operations on discrete sequences.

3.1 Dataset and Task Definition

We consider a dataset $\mathcal{D} = \{(x^\mu, y^\mu)\}_{\mu=1}^n$ where inputs are sequences of length T composed of discrete tokens drawn from a vocabulary $\mathcal{L}$ of size L.

The Histogram Task. The objective is to predict the frequency of occurrence of each token within the sequence. For a given input sequence $x^\mu = [x_1^\mu, \ldots, x_T^\mu]$, the target label $y^\mu \in \mathbb{R}^T$ is defined such that each component y_i^μ contains the total count of the token x_i^μ in the sequence:

$$y_i^\mu = \sum_{j=1}^{T} \mathbf{1}(x_i^\mu = x_j^\mu) \tag{10}$$

where $\mathbf{1}(\cdot)$ is the indicator function. To ensure the dataset covers a wide range of histogram configurations (varying numbers of unique tokens and repetition patterns), we employ a specific controlled random composition procedure. The details of this generation algorithm and the statistical properties of the dataset are detailed in Appendix F.

Illustrative Example. Consider a vocabulary of letters and a sequence of length $T = 7$. The task maps the tokens to their respective counts:

Input (x^μ):	[A,	B,	A,	C,	C,	C,	D]
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
Target (y^μ):	[2,	1,	2,	3,	3,	3,	1]

In this example, 'A' appears twice, 'B' once, 'C' three times, and 'D' once. The model must learn to attend to all positions j where $x_j = x_i$ to compute this sum.

3.2 Student Architecture

The model architecture consists of three stages: a learnable embedding layer, a single-layer ($l = 1$), single-head ($h = 1$) attention mechanism, and a Multi-Layer Perceptron (MLP) readout head.

Embedding. Since the inputs are discrete symbols from a vocabulary of size L, each token $x_t^\mu \in \{0, \ldots, L-1\}$ is first mapped to a continuous vector space using a learnable embedding matrix $E \in \mathbb{R}^{L \times d}$. The input sequence $X^\mu \in \mathbb{R}^{T \times d}$ is given by:

$$X_t^\mu = E_{x_t^\mu}, \quad t = 1, \ldots, T \tag{11}$$

Attention Mechanism. The core interaction is modeled by the matrix $S \in \mathbb{R}^{d\times d}$. As in the synthetic setting, we compare three parameterizations:

$$S = \begin{cases} \frac{1}{\sqrt{rd}} W W^\top & \text{Tied (Factorized)} \\ \frac{1}{\sqrt{rd}} W_Q W_K^\top & \text{Untied} \\ S_{\text{unfac}} & \text{Unfactorized} \end{cases} \tag{12}$$

where the factor matrices $W, W_K, W_Q \in \mathbb{R}^{d\times r}$ consist of i.i.d. standard Gaussian initialization, with the rank fixed at $r = \rho d$. The entries of the full-rank matrix S_{unfac} are drawn independently from $\mathcal{N}(0, 1/d)$ and the entries of W, W_K and W_Q are drawn independently from $\mathcal{N}(0, 1)$.

The context vector z_t^μ for the t-th token is computed by aggregating information from the sequence based on the attention scores h^μ defined in Section 2, Eq. 3 :

$$z_t^\mu = \sum_{j=1}^{T} \left[\sigma_\beta(h^\mu)\right]_{tj} X_j^\mu \tag{13}$$

Readout (MLP). The context vectors are passed through a two-layer MLP to produce the final predictions. The output is a set of logits over the C possible class values (representing the possible counts $\{1, \ldots, T\}$):

$$\text{logits}_t^\mu = W_2 \left(\text{ReLU}(W_1 z_t^\mu + b_1)\right), \quad t = 1, \ldots, T \tag{14}$$

where the dimensions are defined as:

- $W_1 \in \mathbb{R}^{d_{\text{MLP}}\times d}$ and $b_1 \in \mathbb{R}^{d_{\text{MLP}}}$ (Hidden Layer).
- $W_2 \in \mathbb{R}^{C\times d_{\text{MLP}}}$ and $b_2 \in \mathbb{R}^{C}$ fixed to null vector contribution (Output Layer).
- $C = T$: The number of classes corresponds to the maximum possible count frequency.

The final predicted count is obtained by taking the argmax of the logits or by sampling from the resulting categorical distribution.

3.3 Algorithm and Training

The learning process involves optimizing the model parameters $\Theta = \{E, S, W, W_Q, W_K, W_1, b_1, W_2\}$ to map input sequences to the correct token counts.

Training Objective. Since the task is modeled as a multi-class classification problem , its learning dynamics follow high-dimensional asymptotic behaviors [27], while the spectral properties remain universal [28]. Let Θ denote the set of all learnable parameters. Depending on the architecture, Θ includes the embedding matrix E, the attention parameters (S or $\{W, W_Q, W_K\}$), the MLP hidden weights W_1 and bias b_1, and the output weights W_2 (the output bias b_2 is set to zero ($b_2 = 0$) and remains non-learnable). For a batch of n samples, the total training loss is:

$$\mathcal{L}(\Theta) = \frac{1}{n} \sum_{\mu=1}^{n} \sum_{t=1}^{T} \mathcal{L}_{\text{CE}}(\text{logits}_t^\mu, y_t^\mu) + \lambda \sum_{P\in\Theta} \|P\|_F^2 \tag{15}$$

where $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy function [29], and the regularization term applies a global weight decay penalty λ uniformly to all learnable parameters $P \in \Theta$. Training is performed using Adam Optimizer [26] with a learning rate η, continuing until the absolute difference in total loss between consecutive epochs falls below a specified tolerance threshold (tol).

Inference. To obtain the discrete predicted counts $\hat{y}^\mu \in \mathbb{R}^T$, we select the class c with the highest probability mass (argmax) from the output logits:

$$\hat{y}_t^\mu = \arg\max_c [\mathrm{softmax}(\mathrm{logits}_t^\mu)] \tag{16}$$

We model the count prediction as a classification task using Cross-Entropy loss. While this loss is permutation-invariant, the additive structure of the attention mechanism provides a strong inductive bias: by aggregating embedding vectors, the model is incentivized to organize its latent space such that numerical proximity correlates with vector similarity, effectively bridging the gap between classification and the underlying regression task.

Evaluation Metrics. We assess performance using two complementary metrics on the held-out test set composed of n_{test} samples:

1. **Sequence Accuracy:** The fraction of sequences where the model correctly predicts the count for every token (Exact Match).

$$\mathrm{accuracy} = \frac{1}{n_{\mathrm{test}}} \sum_{\mu=1}^{n_{\mathrm{test}}} \mathbf{1}\left(y_\mu^\star = \hat{y}_\mu\right) \tag{17}$$

2. **Generalization Error (MSE):** To quantify how far the predicted counts are from the truth (even if incorrect), we compute the Mean Squared Error on the integer values:

$$\varepsilon_{\mathrm{gen}} = \frac{1}{n_{\mathrm{test}}} \sum_{\mu=1}^{n_{\mathrm{test}}} \|y_\mu^\star - \hat{y}_\mu\|_2^2 \tag{18}$$

3.4 Results

Figure 2 summarizes the performance of the different attention architectures on the Histogram task. We report both the Sequence Accuracy (probability of predicting the entire count sequence correctly) and the Label Error (MSE) as functions of the sample complexity α. Note that we report performance as a function of $\alpha = n/d^2$ rather than the vocabulary size L. Since the attention scores are computed in the embedding space, the learning dynamics are primarily governed by the capacity of the interaction matrix $S \in \mathbb{R}^{d\times d}$, making d^2 the effective scaling dimension.

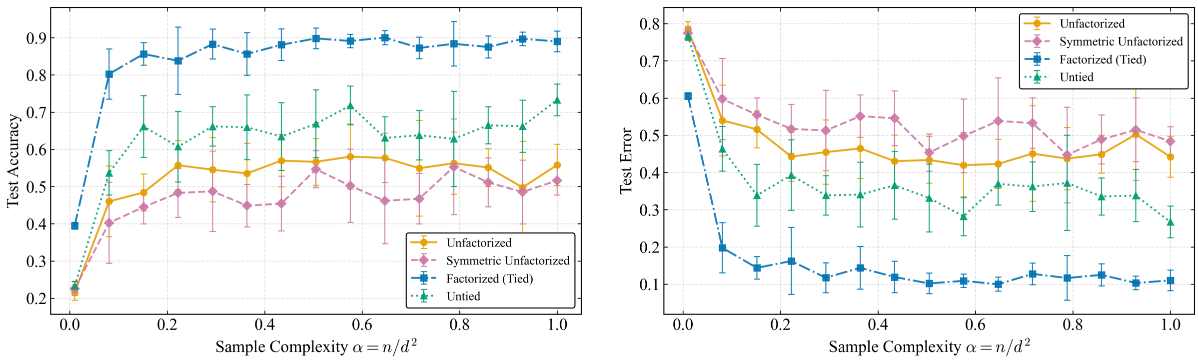
Performance Hierarchy. We observe a clear hierarchy in generalization capability, consistent with our findings on the synthetic dataset:

1. **Tied Attention (Best):** The Tied model achieves high accuracy ($\sim 90\%$) even at low sample complexity ($\alpha \leq 1$). By enforcing both the PSD structure and the zero-diagonal constraint, it efficiently learns the semantic similarity required for counting. Notably, the observed accuracy plateau at 0.9 likely stems from a systematic difficulty in resolving diagonal attention components, as visualizations in Appendix H reveal a persistent attenuation of self-interaction scores compared to off-diagonal semantic blocks. This Tied architecture exhibits low variance across seeds, confirming its robustness.

2. **Untied Attention (Intermediate):** The Untied model (green curve) fails to reach the near-perfect accuracy of the Tied model, plateauing at approximately 70%. The absence of an explicit PSD constraint results in significantly lower data efficiency compared to the

Tied architecture. Notably, we observe a spontaneous reduction of the numerical rank in the factor matrices W_Q and W_K during training. This suggests that the model implicitly discovers a low-dimensional representation despite its full-rank initialization, proving that the low-rank structure is an emergent property of the factorization rather than a hard architectural constraint. A detailed spectral analysis and further evidence of this rank reduction are provided in Appendix I.

3. **Unfactorized Models (Worst):** Both standard and Symmetric Unfactorized models fail to generalize. This is not an optimization failure but a clear case of severe overfitting: we verified that the training loss converges to near-zero ($< 10^{-4}$), confirming that the models successfully interpolate the training data using their excessive degrees of freedom (d^2) without recovering the semantic rule (see Appendices J and K). Crucially, unlike factorized architectures, these models maintain their initial full numerical rank throughout the entire optimization process. As illustrated in Figure 13 placed in Appendix I, this lack of spontaneous rank reduction highlights the absence of implicit regularization, leaving the models trapped in a high-dimensional state that prioritizes noise interpolation over structural learning.



(a) Test Accuracy vs. α (b) Label Error (MSE) vs. α

Figure 2: **Performance Comparison on the Histogram Task.** Panels show (a) Sequence Test Accuracy (higher is better) and (b) Label Mean Squared Error (lower is better) as functions of sample complexity $\alpha = n/d^2$. **Experimental Settings:** Student network with embedding $d = 100$, vocabulary $L = 15$, sequence length $T = 30$, MLP hidden dimension $d_{\text{MLP}} = 100$, and $\beta = 1$. Models are trained for a maximum of 20 000 iterations. Optimization via Adam ($\eta = 0.01$, tol $= 10^{-5}$) with L_2 regularization λ selected via cross-validation in $\{10^{-5}, \dots, 10^{-1}\}$. We evaluate over $n_{\text{test}} = 2\,000$ test samples. **Architectures:** Tied ($\rho = 1.0$), Untied ($\rho = 1.0$), Unfactorized, and Symmetric Unfactorized. Results are averaged over 10 independent training runs performed on a fixed dataset (generated with a fixed seed). This protocol isolates the variance arising purely from weight initialization and SGD dynamics, allowing us to assess the optimization stability of each architecture independent of data sampling noise.

Implicit Regularization vs. Parameter Count. As summarized in Table 1, a counter-intuitive finding emerges regarding model complexity. In the $\rho = 1$ setting, the Untied model possesses the largest number of learnable parameters ($2d^2$), yet it generalizes significantly better than the Unfactorized model (d^2). This confirms that the performance gap is not driven by parameter parsimony, but by the implicit inductive bias of the factorization $W_Q W_K^\top$, which favors low-rank solutions during optimization. Conversely, the Symmetric Unfactorized model has the fewest parameters ($\approx 0.5d^2$) among full-rank variants but performs poorly, demonstrating that parameter reduction alone (without the specific PSD geometry) is insufficient for this task.

Table 1: **Comparison of Model Complexity.** Number of learnable parameters in the attention mechanism for dimension d and rank r.

Architecture	**Parameters**	**Count (for $\rho = 1$, $d \gg 1$)**
Tied ($WW^\top$)	dr	d^2
Untied ($W_Q W_K^\top$)	$2dr$	$2d^2$
Unfactorized (S)	d^2	d^2
Sym. Unfactorized ($S = S^\top$)	$d(d+1)/2$	$\approx 0.5d^2$

Impact of Inductive Bias. Furthermore, the comparison between the Symmetric Unfactorized model and the Tied model highlights that symmetry alone is insufficient. Although the Symmetric model has the capacity to represent the PSD solution, we observe that it fails to eliminate negative eigenvalues during training (see Appendix K). The PSD constraint is therefore essential not just for initialization, but to prevent the optimization from stagnating in saddle points characterized by non-physical repulsive interactions.

4 Conclusion

In this work, we have investigated the foundational mechanisms of sequence modeling by analyzing the learning dynamics of attention layers in high-dimensional regimes. By combining a theoretically tractable synthetic framework with a structured discrete histogram task, we successfully isolated the spectral signatures of architectural inductive biases in Transformers.

Synthetic Findings. Our high-dimensional analysis establishes a clear hierarchy of inductive biases governed by architectural geometry. We demonstrate that while asymmetric architectures (both factorized and full-rank) exhibit a synchronized generalization peak at $\alpha \in [0.5, 0.6]$, the Positive Semi-Definite (PSD) constraint inherent to the Tied model significantly accelerates learning, shifting the interpolation threshold to $\alpha \approx 0.2$. Furthermore, the persistent performance gap between Untied and Unfactorized models in the low-data regime confirms that factorization acts as a powerful structural prior, favoring generalizable low-rank solutions even when theoretical capacity is not a bottleneck. These results suggest that the "spectral signature" of an architecture is a more reliable predictor of data efficiency than raw parameter count.

The Geometry of Attention. Our investigation on the structured discrete counting task confirms a strict performance hierarchy: Tied $>$ Untied $\gg$ Unfactorized. Crucially, we show that symmetry alone is insufficient for generalization; instead, the PSD geometry enforced by the $WW^\top$ factorization acts as a vital regularizer that guides optimization towards semantic similarity while suppressing unstructured noise. This hierarchy is fundamentally driven by the spontaneous reduction of numerical rank observed in factorized architectures. Even with full-rank capacity, Tied and Untied models synchronously compress their representations to match the task's intrinsic dimensionality, whereas Unfactorized models remain stuck in a high-rank state, prioritizing noise interpolation over rule recovery.

Implications and Future Directions. These findings underscore that the Q, K factorization in Transformers is not merely a computational convenience but a structural necessity for data-efficient learning. Building on the fundamental limits established in [30], future work should extend this analysis to deep architectures ($l \gg 1$) to understand how these spectral properties propagate across stacked layers. To better align with modern large-scale models, a key next step involves implementing standard architectures, such as GPT-2, to study these behaviors in more complex environments while replacing the MLP readout with alternative heads. While we identify the 'Saliency Trap' as the dominant failure mode for unstructured parameterizations, the precise topology of the optimization landscape warrants further investigation to characterize the basins of attraction of these competing minima. Finally, exploring learning dynamics for larger vocabularies ($L \sim \mathcal{O}(d)$) will be essential to identify the scaling limits of semantic attention in discrete latent spaces.

5 Acknowledgements

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A Moments and Variance of Quadratic Forms

We compute the expectation and variance of quadratic forms of the type $xSx^\top$ for Gaussian inputs. This derivation is crucial for understanding the signal propagation and the stability of the attention mechanism across different parameterizations (Tied, Untied, Unfactorized).

A.1 Setup

Let $x \in \mathbb{R}^{T\times d}$ be a random matrix representing the input sequence, where the rows $x_a \in \mathbb{R}^d$ $(a = 1, \dots, T)$ are independent standard Gaussian vectors (as presented in Eq. 2). We consider the attention score matrix A (before softmax) defined by the quadratic form:

$$A_{ab} = \frac{1}{\sqrt{d}}(xSx^\top)_{ab} = \frac{1}{\sqrt{d}}x_a^\top S x_b \tag{19}$$

In the following, we analyze the statistics of the unnormalized quadratic form $Q_{ab} = x_a^\top S x_b$.

A.2 Expectation

Since x_a is centered, linearity of expectation gives:

$$\mathbb{E}[x_a^\top S x_b] = \mathrm{Tr}\left(S\,\mathbb{E}[x_b x_a^\top]\right)$$

Using the independence $\mathbb{E}[x_b x_a^\top] = \delta_{ab} I_d$, we obtain:

$$\boxed{\mathbb{E}[x_a^\top S x_b] = \delta_{ab}\,\mathrm{Tr}(S)} \tag{20}$$

Note that for the Untied and Unfactorized models initialized with centered random weights, $\mathbb{E}[\mathrm{Tr}(S)] = 0$. For the Tied model $(S = WW^\top)$, $\mathrm{Tr}(S) > 0$, necessitating the trace correction term discussed in the main text.

A.3 Variance

We compute the variance $\mathrm{Var}(Q_{ab}) = \mathbb{E}[Q_{ab}^2] - (\mathbb{E}[Q_{ab}])^2$.

Case 1: Off-diagonal terms ($a \neq b$). Since x_a and x_b are independent:

$$\mathbb{E}[(x_a^\top S x_b)^2] = \mathbb{E}\left[\sum_{i,j,k,l} x_{ai} S_{ij} x_{bj} x_{ak} S_{kl} x_{bl}\right]$$

Using independence, $\mathbb{E}[x_{ai}x_{ak}] = \delta_{ik}$ and $\mathbb{E}[x_{bj}x_{bl}] = \delta_{jl}$. The sum becomes:

$$\sum_{i,j,k,l} S_{ij} S_{kl} \delta_{ik} \delta_{jl} = \sum_{i,j} S_{ij} S_{ij} = \mathrm{Tr}(SS^\top) = \|S\|_F^2$$

Thus, for $a \neq b$:

$$\boxed{\mathrm{Var}(x_a^\top S x_b) = \mathrm{Tr}(SS^\top)} \tag{21}$$

Case 2: Diagonal terms ($a = b$). Let $x = x_a$. We compute the second moment of $x^\top Sx$:

$$\mathbb{E}[(x^\top Sx)^2] = \sum_{i,j,k,l} S_{ij}S_{kl}\mathbb{E}[x_i x_j x_k x_l]$$

Using Wick's theorem (Isserlis), the fourth moment expands into three terms:

$$\mathbb{E}[x_i x_j x_k x_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$$

Substituting back:

- Term 1 ($\delta_{ij}\delta_{kl}$): $\sum S_{ii}S_{kk} = (\operatorname{Tr} S)^2$
- Term 2 ($\delta_{ik}\delta_{jl}$): $\sum S_{ij}S_{ij} = \operatorname{Tr}(SS^\top)$
- Term 3 ($\delta_{il}\delta_{jk}$): $\sum S_{ij}S_{ji} = \operatorname{Tr}(S^2)$

Subtracting the squared mean $(\mathbb{E}[Q])^2 = (\operatorname{Tr} S)^2$ (Term 1), we find:

$$\boxed{\operatorname{Var}(x_a^\top Sx_a) = \operatorname{Tr}(SS^\top) + \operatorname{Tr}(S^2)} \tag{22}$$

Implication for Symmetric vs. Non-Symmetric S.

- For **Tied** attention ($S = S^\top$): $\operatorname{Tr}(S^2) = \operatorname{Tr}(SS^\top)$. The variance of the diagonal is exactly twice the variance of the off-diagonal terms ($2\|S\|_F^2$).
- For **Unfactorized** attention ($S \neq S^\top$): The term $\operatorname{Tr}(S^2)$ can be small or negative. The variance is dominated by the Frobenius norm $\|S\|_F^2$.

A.4 Spectral Scaling of the Unfactorized Model

We initialize the matrix $S \in \mathbb{R}^{d\times d}$ with i.i.d. Gaussian entries as presented in Eq. 8. In this regime, S belongs to the real Ginibre ensemble (non-symmetric random matrices). Unlike Wigner matrices (symmetric) which follow a semicircle law on the real line, the eigenvalues of S follow the Circular Law.

As $d \to \infty$, the eigenvalues are uniformly distributed in the complex plane on the disk of radius:

$$R = \sigma\sqrt{d} = \frac{1}{\sqrt{d}}\sqrt{d} = 1 \tag{23}$$

This confirms that the spectral radius is $O(1)$. This scaling choice $\sigma = 1/\sqrt{d}$ is critical: it prevents the spectrum from exploding (which would occur if $\sigma = 1$, giving radius $\sqrt{d}$) or collapsing, ensuring stable gradient propagation through the attention mechanism.

B Validation: Tied Attention in the Realizable Setting

As a validation step for our experimental framework, we reproduce the learning dynamics reported in [14] for the single-layer ($l = 1$) tied-attention model. We place ourselves in the realizable setting, where the student model matches the teacher architecture exactly, serving as a controlled benchmark.

B.1 Setup

Model Parameterization. The attention kernel is parameterized as a factored symmetric matrix:

$$S = \frac{WW^\top}{\sqrt{rd}}, \qquad W \in \mathbb{R}^{d\times r}, \qquad W_{ij} \sim \mathcal{N}(0,1) \tag{24}$$

where $r = \rho d$ is controlled by the width ratio $\rho > 0$. For a pair of tokens $x_a, x_b \in \mathbb{R}^d$, the model output is given by:

$$y_{ab} = \sigma_\beta\left(\frac{x_a^\top S x_b - \delta_{ab}\,\mathrm{Tr}(S)}{\sqrt{d}}\right). \tag{25}$$

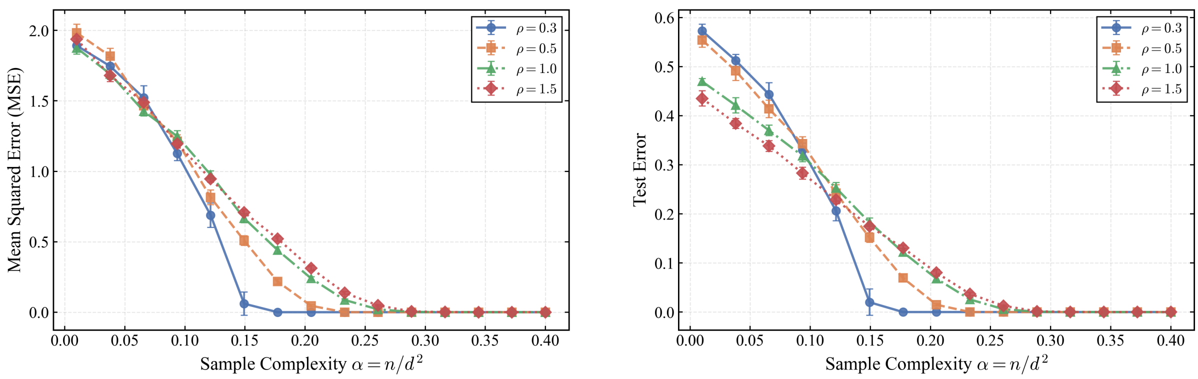
Teacher-Student Setting. The target labels $y^\star_\mu$ are generated using a ground-truth matrix $S^\star$ constructed with the same rank ratio ρ as the student. We assume noiseless activations $(\Delta = 0)$ and standard Gaussian inputs $x_\mu \sim \mathcal{N}(0, \mathbb{I}_{T\times d})$ with sequence length T and embedding dimension d.

Learning Objective. We perform Empirical Risk Minimization (ERM). Student parameters W are optimized to minimize the Sum of Squared Errors (SSE) between the teacher and student probability distributions.

$$\mathcal{L}(W) = \sum_{\mu=1}^{n} \left\| y^\star_\mu - \hat{y}_\mu(W) \right\|_F^2 \tag{26}$$

Optimization is performed using Adam [26] without weight decay regularization.

B.2 Results



(a) Reconstruction MSE ($\mathcal{E}_{\mathrm{MSE}}$) (b) Generalization Error ($\mathcal{E}_{\mathrm{gen}}$)

Figure 3: **Validation on Tied-Attention (Realizable Setting).** Reconstruction error (left) and generalization error (right) versus sample complexity $\alpha = n/d^2$ for varying width ratios $\rho \in \{0.3, 0.5, 1.0, 1.5\}$. **Setup:** $d = 100$, $T = 2$, $\beta = 1$, $\Delta = 0$, $n_{\mathrm{test}} = 2\,000$. Training via Adam ($\eta = 0.1$, tol $= 10^{-6}$) averaged over 16 experiments. **Observation:** Errors decrease monotonically with α. Models with higher ρ possess more degrees of freedom, requiring higher sample complexity to converge.

We evaluate the performance using two metrics: the Reconstruction Error (estimation of the matrix $S^\star$) and the Generalization Error :

$$\mathcal{E}_{\mathrm{MSE}}(\hat{S}) = \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}_{\mathrm{test}}}\left[\frac{1}{d}\left\|\hat{S} - S^\star\right\|_F^2\right], \qquad \mathcal{E}_{\mathrm{gen}}(\hat{y}) = \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}_{\mathrm{test}}}\left[\left\|y - \hat{y}(x,\hat{S})\right\|_F^2\right] \tag{27}$$

Figure 3 displays these metrics as functions of the sample complexity $\alpha = n/d^2$.

Analysis. Our experiments reproduce the qualitative behavior reported in the literature [14]. We observe a monotonic decrease in both MSE and label error as α increases. Crucially, the learning difficulty scales with the width ratio ρ: even in the over-parameterized regime ($\rho = 1.5$), the symmetric factorization provides a strong inductive bias that allows smooth convergence, validating our training pipeline for the subsequent analyses.

C Validation: Tied Attention under Noisy ERM

As a second validation step, we extend the reproduction of [15] to the Empirical Risk Minimization (ERM) setting with noisy labels. Unlike the Bayes-optimal case, here the student does not know the noise level, and the training objective includes explicit regularization.

C.1 Setup

Model Parameterization. We consider a single-layer ($l = 1$) tied-attention model. The student and teacher kernels are parameterized as:

$$S = \frac{WW^\top}{\sqrt{rd}}, \qquad S^\star = \frac{W^\star {W^\star}^\top}{\sqrt{r^\star d}} \tag{28}$$

with embedding dimension $d = 100$. The width of the matrices are defined by the width ratios $\rho = r/d$ (student) and $\rho^\star = r^\star/d$ (teacher). In this experiment, we set $(\rho, \rho^\star) = (1.0, 0.5)$. This corresponds to an over-parameterized regime where the student has full-rank capacity ($\rho = 1$) attempting to learn a low-rank teacher ($\rho^\star = 0.5$).

Noisy Teacher and ERM Objective. The teacher scores are corrupted with additive Gaussian noise of amplitude $\Delta = 0.5$. The student minimizes the regularized Mean Squared Error (Ridge Regression):

$$\mathcal{L}(W) = \frac{1}{n}\sum_{\mu=1}^{n}\left\|y^{\mu^\star} - \hat{y}^\mu(W)\right\|_F^2 + \lambda\|W\|_F^2 \tag{29}$$

where λ is the weight decay coefficient.

Performance Metrics. We quantify the Generalization Error ($\mathcal{E}_{\mathrm{gen}}$) as functions of the sample complexity $\alpha = n/d^2$:

$$\mathcal{E}_{\mathrm{gen}}(\hat{y}) = \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}_{\mathrm{test}}}\left[\|y - \hat{y}(x,\hat{W})\|_F^2\right] \tag{30}$$

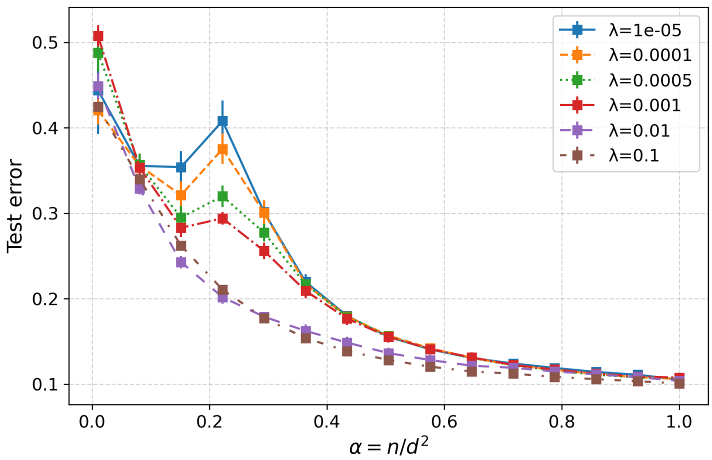
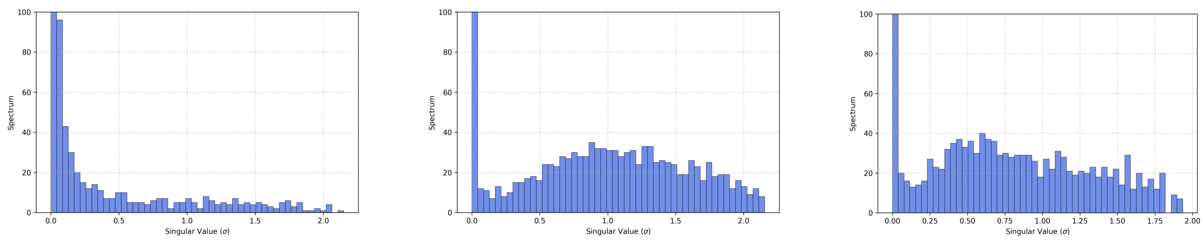


Figure 4: **Generalization Error under Noisy Supervision.** Test error plotted against sample complexity α for varying regularization strengths λ. Unlike the monotonic Bayes-optimal curves, we observe a distinct **generalization peak** around $\alpha \approx 0.2$. This peak is amplified as regularization decreases (smaller λ), characteristic of the double-descent phenomenon in noisy ERM. **Settings:** $T = 2$ tokens, $\beta = 1$, $\Delta = 0.5$, $\rho = 1.0$, $\rho^\star = 0.5$, $n_{\text{test}} = 2\ 000$, optimized via Adam ($\eta = 0.1$, tol $= 10^{-6}$). We scan the regularization strength λ over the set $\{10^{-1}, 10^{-2}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}, 10^{-5}\}$. Results are averaged over 16 independent experiments, and test metrics are evaluated on 2 000 samples.



(a) $\alpha = 0.01$ (b) $\alpha = 0.36$ (c) $\alpha = 1.0$

Figure 5: **Spectral Dynamics.** Cumulative singular value distribution of the learned attention matrix S for sample complexity $\alpha \in \{0.01, 0.36, 1.0\}$. At low sample complexity (a), the spectrum is almost random and is governed only by a delta peak at zero. As α increases (c), the spectrum exhibits a separation between signal eigenvalues (matching the teacher) and noise.

C.2 Results

Summary of Findings. The introduction of label noise ($\Delta = 0.5$) reveals a non-monotonic generalization landscape, contrasting with the monotonic decay typical of Bayes-optimal settings. We observe a distinct generalization peak (interpolation cusp) around $\alpha \approx 0.2$ under weak regularization. This early peak, occurring well before the nominal full-rank capacity ($\alpha = 1$), underscores the drastic reduction in effective degrees of freedom induced by the Tied PSD constraint. Spectral analysis (Fig. 5) confirms a phase transition: at low sample complexity, the student's spectrum is dominated by a noise bulk, while for $\alpha \geq 0.36$, informative eigenvalues emerge from the bulk to align with the teacher's low-rank subspace ($\rho^\star = 0.5$). Increasing regularization λ effectively smooths this transition, acting as a spectral filter that suppresses noise-driven eigenvalues and guides the optimizer toward the semantic solution.

D Untied Attention: Robustness to Over-Parameterization

In this appendix, we investigate the sensitivity of the Untied attention mechanism to its capacity. Specifically, we fix the complexity of the Teacher model at $\rho^\star = 0.5$ and vary the width ratio of the Student model $\rho = r/d$. This allows us to study the regime of over-parameterization where the student's capacity strictly exceeds the teacher's rank ($\rho > \rho^\star$). We consider four distinct student configurations: $\rho \in \{0.5, 0.75, 1.0, 1.5\}$.

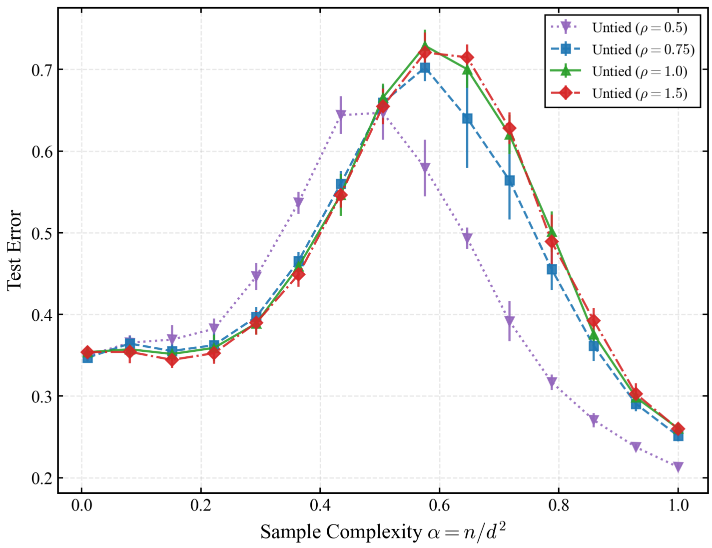


Figure 6: **Impact of Student Width Ratio (ρ) on Generalization.** Test error as a function of sample complexity α for varying student width ratios $\rho \in \{0.5, 0.75, 1.0, 1.5\}$. The Teacher complexity is fixed at $\rho^\star = 0.5$. **Settings:** $T = 2$ tokens, $\beta = 1$, $\lambda = 10^{-4}$, $\Delta = 0.5$, optimized via Adam ($\eta = 0.1$, tol $= 10^{-6}$) averaged over 16 experiments.

The results in Figure 6 reveal two critical insights regarding model capacity. First, we observe a clear saturation of performance once the student capacity matches the teacher complexity ($\rho \geq 0.5$), as the learning curves for $\rho \in \{0.75, 1.0, 1.5\}$ largely overlap with the matched case. Second, and most crucially, the model exhibits a remarkable robustness to over-parameterization. Even when the student width ratio is increased to $\rho = 1.5$ (resulting in a model with $1.5 \times d^2$ parameters, 1.5 times more than the Unfactorized baseline) performance does not degrade. This successful generalization stands in stark contrast to the failure of the Unfactorized model (see Appendix E) and confirms that the explicit factorization $S = W_Q W_K^\top$ acts as a powerful implicit regularizer. By biasing the optimization towards low-rank solutions, the Untied architecture avoids the severe overfitting typically associated with such massive parameter counts in high-dimensional regression.

E Unfactorized Model Analysis on Synthetic Data

In this appendix, we analyze the learning dynamics of the Unfactorized attention model in the matched setting. Here, both the Student S and the Teacher $S^\star$ are full-rank matrices initialized with i.i.d. Gaussian entries, without any symmetry constraint. This analysis follows the same mathematical framework as the synthetic Teacher-Student model established in the main report.

E.1 Generalization Error

We analyze the generalization performance (Label Error) as a function of the sample complexity $\alpha = n/d^2$.

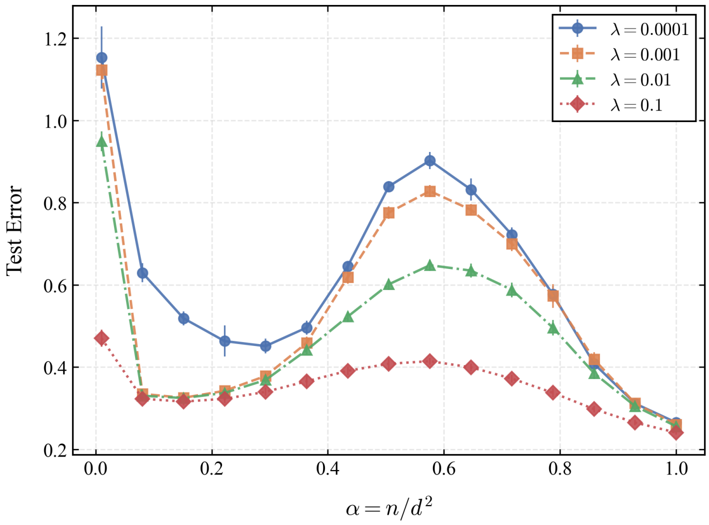


Figure 7: **Generalization Error of the Unfactorized Model.** Test error versus sample complexity α for varying regularization strengths. We observe a characteristic peak in the error around $\alpha \approx 0.5$. This non-monotonic behavior is indicative of the "double descent" phenomenon in the noisy regime: the model overfits the noise when the effective capacity is comparable to the sample size. **Settings:** $T = 2$ tokens, $\beta = 1$, $\Delta = 0.5$, optimized via Adam ($\eta = 0.1$, tol $= 10^{-6}$) averaged over 16 experiments.

The generalization error (Fig. 7) exhibits a distinct interpolation peak around $\alpha \approx 0.6$, a hallmark of the double descent phenomenon in the noisy regime. We observe that the magnitude of this peak increases significantly as the regularization strength λ decreases, confirming that the model overfits label noise when the effective capacity matches the sample size.

E.2 Spectral Dynamics

To understand the internal mechanism, we analyze the full spectral properties of the learned matrix S. Since the matrix is non-symmetric, its eigenvalues are complex-valued. We compute the full spectrum and analyze the distribution of their real parts. For a random Gaussian matrix (like the Teacher $S^\star$, which belongs to the Real Ginibre Ensemble), the eigenvalues are distributed uniformly on a disk in the complex plane (the Circular Law). Mathematically, the projection of this uniform disk onto the real axis follows a semi-circular density. We track how the Student's spectrum evolves towards this distribution. Figure 8 illustrates this evolution:

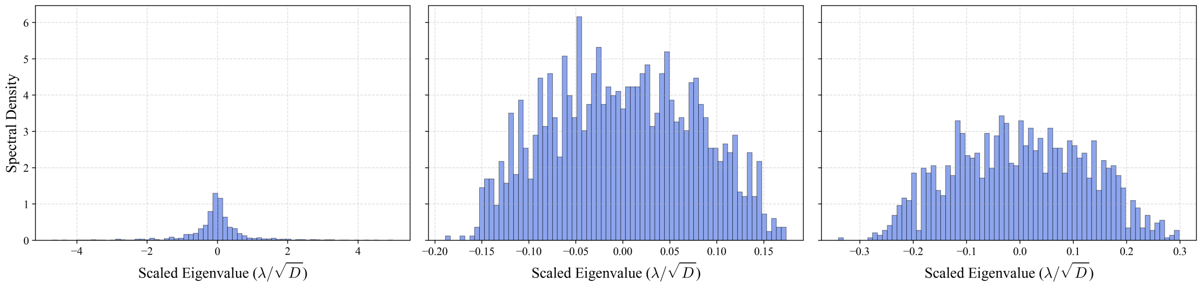


Figure 8: **Spectral Evolution of the Unfactorized Matrix.** Histograms of the real part of the eigenvalues for fixed regularization $\lambda = 10^{-4}$ and increasing sample complexity $\alpha \in \{0.01, 0.5, 1.0\}$ (from left to right). **(Left, $\alpha \approx 0$):** The spectrum is collapsed with a sharp peak at zero, due to initialization and regularization. **(Middle, $\alpha \approx 0.5$):** We observe a semi-circular distribution. This corresponds to the projection of the Circular Law onto the real axis, indicating that the Student behaves like a random Gaussian matrix (fitting the noise) in the interpolation regime. **(Right, $\alpha \approx 1.0$):** The distribution stabilizes into a well-defined bulk. This confirms that the Student has converged to the spectral scale of the Teacher $S^\star$.

F Histogram Task Details and Attention Analysis

In this appendix, we provide additional details on the Histogram task generation and analyze the internal representations learned by the model.

F.1 Controlled Data Generation

To ensure that the model learns a robust counting algorithm rather than exploiting statistical artifacts, we do not sample tokens uniformly. If tokens were drawn uniformly from a large vocabulary ($L \gg T$), most sequences would consist of unique tokens (count $= 1$), reducing the task to a trivial identity mapping. Instead, we enforce a rich distribution of counts using the following controlled composition procedure:

1. **Integer Partitioning:** We select a random number of unique tokens $k \in [1, \min(T, L)]$. The total sequence length T is partitioned into k positive integers $c_1, \ldots, c_k$ such that $\sum_{i=1}^{k} c_i = T$. This step determines the counts (e.g., one token appears 3 times, another 1 time).

2. **Token Assignment:** These counts are assigned to k distinct tokens chosen randomly from the vocabulary $\mathcal{L}$.

3. **Shuffling:** The tokens are shuffled to produce the final input sequence x^μ. This ensures that position carries no information about the count.

F.2 Semantic vs. Positional Solutions

The histogram task theoretically admits two types of solutions:

- **Positional Solution:** The model memorizes that specific positions map to specific outputs. This fails to generalize to unseen sequences or permuted inputs.

- **Semantic Solution:** The model computes the output by attending to tokens based on their content similarity.

As discussed in [16], robust generalization requires converging to the semantic solution. Mathematically, the learned attention matrix A should approximate the *Gram matrix* of the token identities:

$$A_{ij}^{\text{ideal}} \propto \mathbf{1}(x_i = x_j) \tag{31}$$

This means identical tokens should attend strongly to each other, regardless of their relative positions in the sequence.

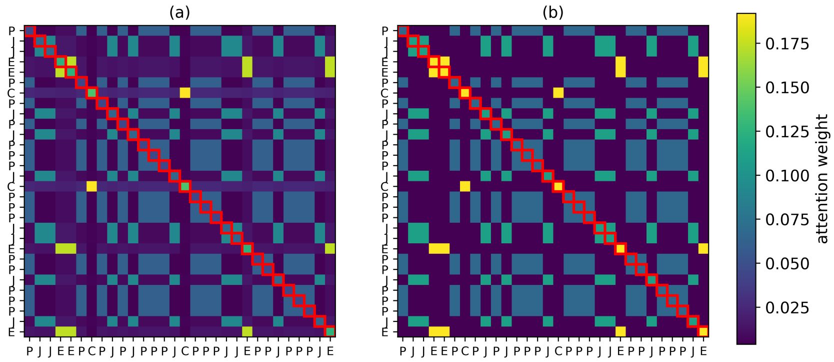


Figure 9: **Visualizing the Semantic Mechanism (Tied Model).** Comparison between the learned attention matrix (Left) and the ground-truth token similarity matrix (Right) for a sample sequence. Darker regions indicate higher attention weights. We observe a strong alignment between the two matrices: the attention weights A_{ij} are significantly larger when $x_i = x_j$. This confirms that the Tied architecture has successfully learned the semantic solution, relying on the positive semi-definite structure of $S = WW^\top$ to capture token similarity rather than memorizing positional patterns.

Figure 9 confirms this hypothesis for our Tied architecture. The learned attention map effectively recovers the ground-truth similarity structure. This visualization highlights why the Tied model generalizes better in the low-sample regime: its PSD constraint ($S = WW^\top$) naturally biases the model towards learning dot-product similarities (semantic matching), whereas Unfactorized models must learn to suppress negative eigenvalues to approximate this behavior.

G Histogram Dataset: Hyperparameter Selection

In this appendix, we justify the choice of the sequence length $T = 30$ and vocabulary size $L = 15$ used in our experiments. While our generation procedure (detailed in Appendix F) explicitly controls the count distribution to ensure diversity, it is crucial to select T and L such that the task naturally resides in a "dense" regime where token repetitions are statistically significant.

G.1 Statistical Justification for T and L

If tokens were sampled uniformly from the vocabulary $\mathcal{L}$ with replacement, the number of occurrences X of a specific token in a sequence of length T would follow a Binomial distribution:

$$X \sim \text{Binomial}\left(n = T, p = \frac{1}{L}\right) \tag{32}$$

with expected count $\mathbb{E}[X] = T/L$. For the histogram task to be non-trivial, the sequence must contain multiple repetitions of the same tokens. If $T \ll L$, sequences would consist mostly of unique tokens (counts of 1), reducing the problem to simple identity mapping. Conversely, we aim for a regime where high counts are probable.

We define a hardness criterion: we require that any given token has a significant probability (at least 10%) of appearing at least 4 times in the sequence. Mathematically:

$$P(X \geq 4) = 1 - \sum_{k=0}^{3} \binom{T}{k} \left(\frac{1}{L}\right)^k \left(1 - \frac{1}{L}\right)^{T-k} \geq 0.1 \tag{33}$$

G.2 Numerical Selection

Figure 10 plots this probability as a function of T for various vocabulary sizes L.

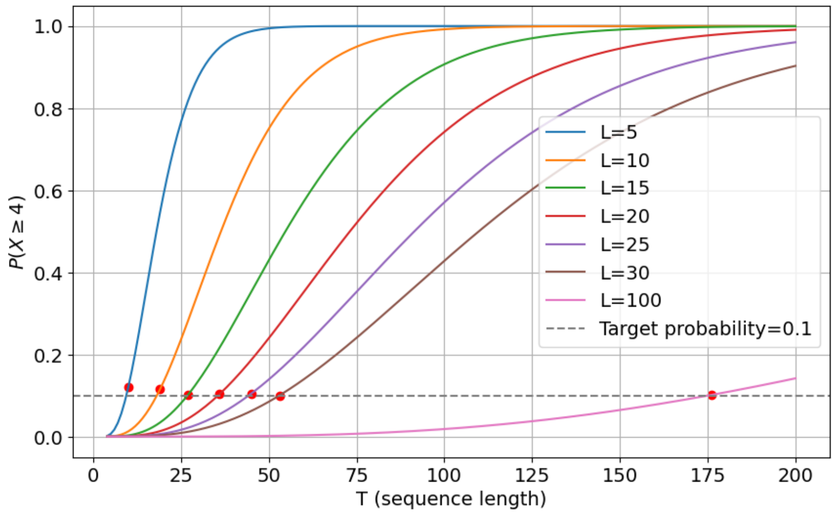


Figure 10: **Regime Selection.** Probability of observing at least four occurrences of a specific token ($P(X \geq 4)$) as a function of sequence length T, for varying vocabulary sizes L. The dashed line represents the 10% threshold.

For our chosen vocabulary size $L = 15$, numerical evaluation shows that a minimum length of $T \approx 27$ is required to satisfy the condition $P(X \geq 4) \geq 0.1$. We therefore select $T = 30$ for our experiments. This choice places the task in a dense regime where collisions are frequent, ensuring that the model must learn a genuine counting mechanism rather than sparse pattern matching. Crucially, this ratio $T/L = 2$ is kept constant relative to the scaling used in larger dimensions (e.g., $L \sim \mathcal{O}(d)$), ensuring that the task difficulty remains comparable across settings.

H Detailed Analysis: Tied Attention on Histogram Task

In this appendix, we investigate the internal representations learned by the **Tied (Factorized)** architecture ($S = WW^\top$) on the Histogram task. As this parameterization enforces a Positive Semi-Definite (PSD) structure, the eigenvalues are non-negative and coincide with the singular values. This constraint is central to the model's ability to robustly recover the counting mechanism.

To understand the learning dynamics, Figure 11 combines the spectral evolution (top) and the resulting attention maps (bottom). The spectral analysis tracks the distribution of the singular values of S. Unlike the Unfactorized models where the spectrum is diffuse or contains negative components, here the values are strictly positive. At low sample complexity ($\alpha = 0.01$), the spectrum is largely collapsed near zero due to initialization and regularization. As α increases towards 1.0, we observe the formation of a distinct spectral bulk. This expansion corresponds to the model learning the relevant subspace defined by the token embeddings, effectively capturing the signal variance required for the task.

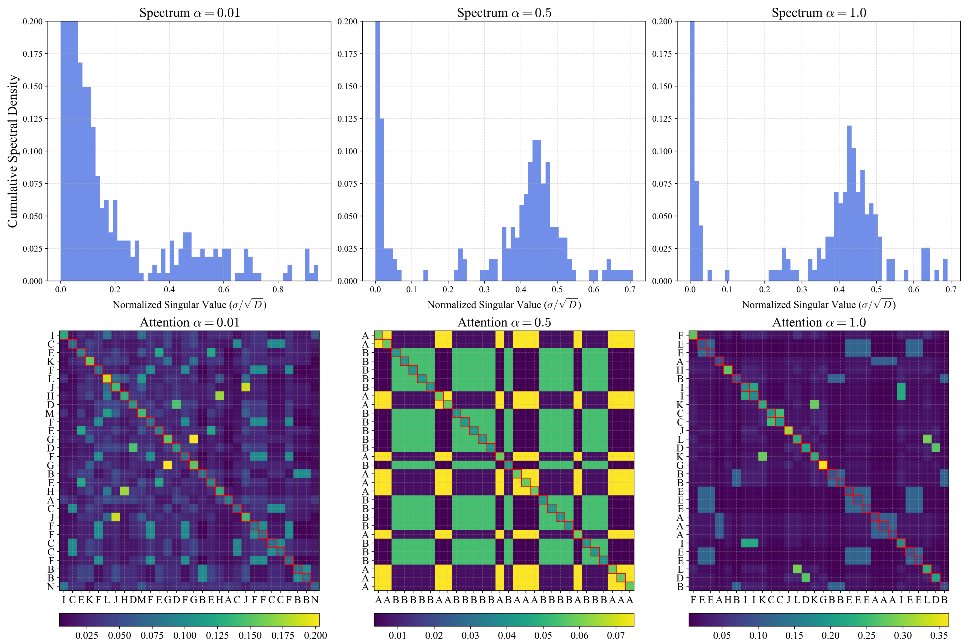


Figure 11: **Learned Representations of Tied Attention. Top:** Evolution of the singular value distribution of S for increasing sample complexities $\alpha \in \{0.01, 0.5, 1.0\}$. We observe the progressive formation of a spectral bulk as the model captures the signal. **Bottom:** Attention scores for a representative test sequence (darker regions indicate higher weights). **Settings:** Student network with $d = 100$, width ratio $\rho = 1$, vocabulary $L = 15$, and sequence length $T = 30$, optimized via Adam ($\eta = 0.01$, tol $= 10^{-5}$).

The corresponding attention maps (bottom panels) confirm that this spectral structure effectively translates into a functional semantic mechanism. The learned attention matrix displays a distinct pattern where significant weights are assigned to pairs of identical tokens ($x_i = x_j$), manifesting as off-diagonal active blocks. This validates that the PSD constraint effectively biases the optimization toward a dot-product similarity metric, enabling the model to aggregate information based on token identity rather than relative position.

The observation that Sequence Accuracy plateaus at approximately 0.9, failing to reach unity even under high sample complexity, may be attributed to a systematic difficulty in resolving the diagonal components. Indeed, the attention visualizations reveal a persistent attenuation of diagonal scores compared to off-diagonal interactions. This suggests that while the optimization successfully recovers relationships between distinct tokens, the self-attention dynamics remain imperfectly resolved, preventing the model from achieving a perfect exact match

I Detailed Analysis: Untied Attention on Histogram Task

We extend our investigation to the Untied (Factorized) architecture, specifically in the regime where the width ratio $\rho = 1.0$ (i.e., $W_Q, W_K \in \mathbb{R}^{d \times d}$). Unlike a bottleneck architecture characterized by $r \ll d$, this model theoretically possesses full-rank capacity and a parameter count of $2d^2$, ($\rho = 1$). The fundamental distinction lies in the factorized structure and the absence of a symmetry constraint ($W_Q \neq W_K$), isolating the effect of the product parameterization on the optimization landscape.

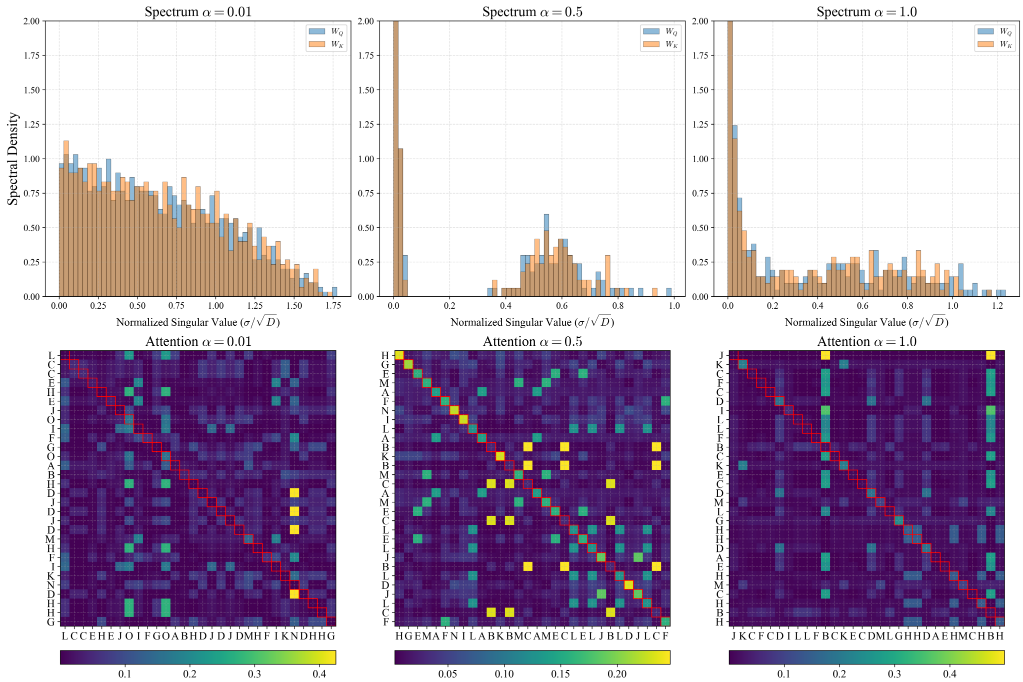


Figure 12: **Learned Representations of Untied Attention. Top:** The singular value distribution shows successful signal extraction from the noise bulk. **Bottom:** The attention maps demonstrate the emergence of a semantic diagonal, though residual vertical artifacts persist due to the lack of a PSD constraint.

Spectral and Functional Variance. Despite its theoretical capacity to overfit high-dimensional noise, the product structure $W_Q W_K^\top$ acts as an implicit regularizer during optimization. As illustrated in Figure 12:

- **Signal Extraction:** The model successfully identifies and amplifies a signal bulk in the singular value spectrum, reflecting the recovery of meaningful token interactions.
- **The Vertical Drop:** However, the resulting attention maps often exhibit a hybrid morphology where semantic diagonal blocks are accompanied by residual vertical "columns". This indicates that without an explicit Positive Semi-Definite (PSD) constraint, the model

remains vulnerable to vertical streak behavior, albeit to a significantly lesser degree than the Unfactorized variants.

Formalization of the Saliency Trap. We characterize the 'Saliency Trap' not merely as a visual artifact of vertical striations, but as a formal spectral collapse of the attention mechanism's relational capacity [13, 9]. In this failure mode, the interaction matrix S fails to approximate the necessary Positive Semi-Definite (PSD) similarity structure required for semantic matching [16, 15]. Empirically, this manifests as a column-wise dominance where specific key tokens capture the majority of the attention weight regardless of the query. Mathematically, the model converges to a rank-1-dominant regime where the attention scores A_{ij} become approximately context-independent [16]. This represents a fundamental collapse of the semantic dimension into a global statistical dimension. The optimizer discovers a "shortcut" by attending to tokens based on their global statistical salience in the training distribution rather than their contextual relevance to the sequence [31]. While this 'Saliency' solution provides a robust attractor in the optimization landscape for unstructured or symmetric-only matrices, it lacks the relational structure necessary for genuine algorithmic generalization [30].

The PSD Constraint as a Barrier to Saliency Collapse. The superiority of the Tied architecture stems from its structural inability to sustain the context-independent modes of the Saliency Trap [14]. In the Tied model, the attention scores are parameterized as a centered quadratic form $h_{ab} \propto x_a^\top (WW^\top) x_b$, which represents a generalized dot-product similarity $\langle W^\top x_a, W^\top x_b \rangle$ in a projected latent space [15]. This Positive Semi-Definite (PSD) geometry ensures that high attention weights are fundamentally coupled to token proximity, effectively "locking" the mechanism into a semantic relational rule [16]. In contrast, the Saliency Trap observed in Unfactorized and Symmetric models requires the optimizer to favor specific key tokens x_j regardless of the query x_i, often by failing to eliminate negative eigenvalues during training [17]. By explicitly forbidding negative eigenvalues and enforcing symmetry through the $WW^\top$ factorization, the Tied architecture eliminates these parasitic minima from the optimization landscape [15]. Consequently, the semantic solution becomes the only robust attractor, explaining why the Tied model achieves high data efficiency while unstructured models remain trapped in non-generalizable, high-rank states [30].

Spontaneous Rank Reduction. A striking phenomenon is observed regarding the effective dimensionality of the learned solutions: although W_Q and W_K are initialized at full rank ($r = d$), they do not maintain it during optimization. As illustrated in Figure 13, both factors undergo a spontaneous and synchronized reduction in numerical rank [32] as training progresses. This implicit bias stems from the synergistic interplay between the factorized parameterization and Gradient Descent dynamics under weight decay. Consistent with established theoretical results , minimizing the L_2 norm of individual factors approximates the minimization of the nuclear norm of the resulting product $S = W_Q W_K^\top$, thereby promoting spectrally sparse, low-rank solutions. The model effectively "discovers" the low-dimensional nature of the task and compresses its representation accordingly, confirming that the observed low-rank structure is an emergent property rather than a hard architectural constraint. Crucially, this reduction is not a trivial consequence of weight decay, but a specific inductive bias inherent to the factorized architecture. This is evidenced by the behavior of the Unfactorized model: even when subjected to identical regularization strengths ($\lambda \in [10^{-4}, 10^{-5}]$), it maintains its initial full numerical rank ($r \approx 100$) and remains trapped in a high-dimensional, overfitted state. This contrast confirms that the product parameterization $W_Q W_K^\top$ acts as a catalyst for spectral sparsity, guiding the optimizer toward a generalizable semantic solution even when explicit regularization is weak.

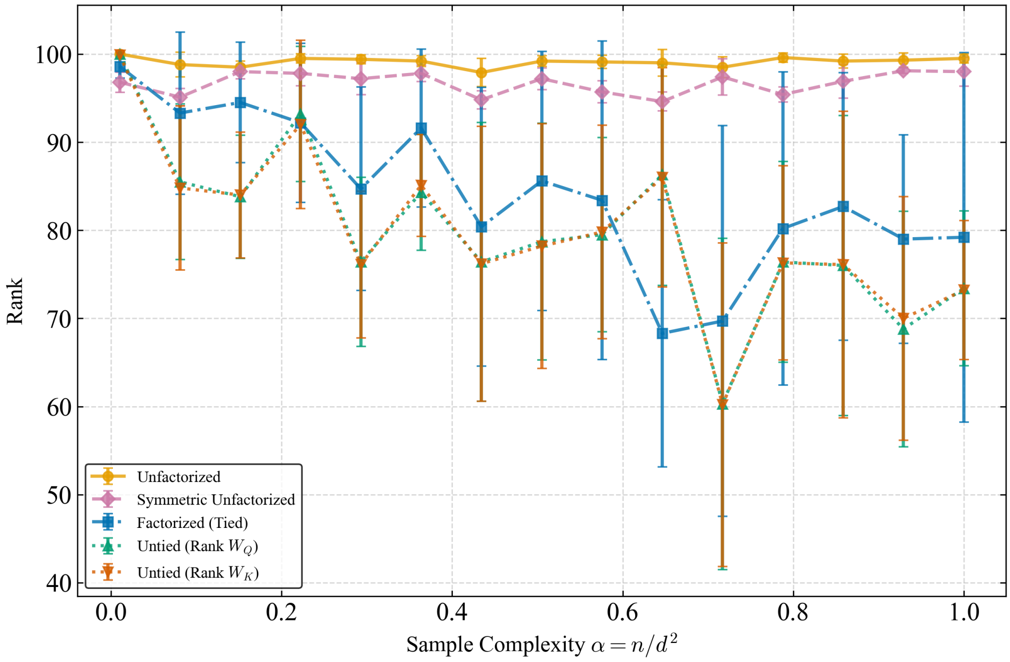


Figure 13: **Spontaneous Rank Reduction and Implicit Regularization.** Evolution of numerical rank across architectures. **Observations:** (i) Unfactorized models consistently preserve their initial full rank ($r \approx 100$), leading to overfitting. (ii) Tied and Untied models exhibit a synchronized and drastic reduction in numerical rank during training. (iii) This reduction confirms that low-rank structure is an emergent property of the factorization.

J Detailed Analysis: Unfactorized Attention on Histogram Task

We analyze the Unfactorized architecture within the sample complexity window $\alpha \leq 1.0$. A key observation in Figure 14 (and the associated error bars in Figure 2) is the high variance of the solution. Unlike the Tied model which consistently converges, the Unfactorized model exhibits a bimodal behavior across the 10 independent training seeds.

Spectral Density vs. Numerical Rank. Regardless of the functional outcome, the eigenvalue spectrum at $\alpha = 1.0$ (Figure 14, top right panel) consistently exhibits a massive concentration of eigenvalues near zero. However, this does not imply a low-rank collapse. As demonstrated in Figure 13, the Unfactorized model maintains its initial full numerical rank ($r \approx 100$) across all values of α. This indicates that while the regularization λ effectively attenuates the magnitude of most eigenmodes, it fails to eliminate them. Consequently, the model still utilizes its full d^2 independent degrees of freedom to interpolate training noise. Unlike factorized architectures, which exhibit spontaneous rank reduction, the Unfactorized model remains trapped in a high-dimensional state where the residual "noise bulk" prevents the recovery of the structural counting rule. The fact that the optimal λ identified via cross-validation is small (10^{-5}) further reinforces the architectural nature of this bias: the low-rank structure emerges from the optimization dynamics themselves rather than being forced by a heavy external penalty.

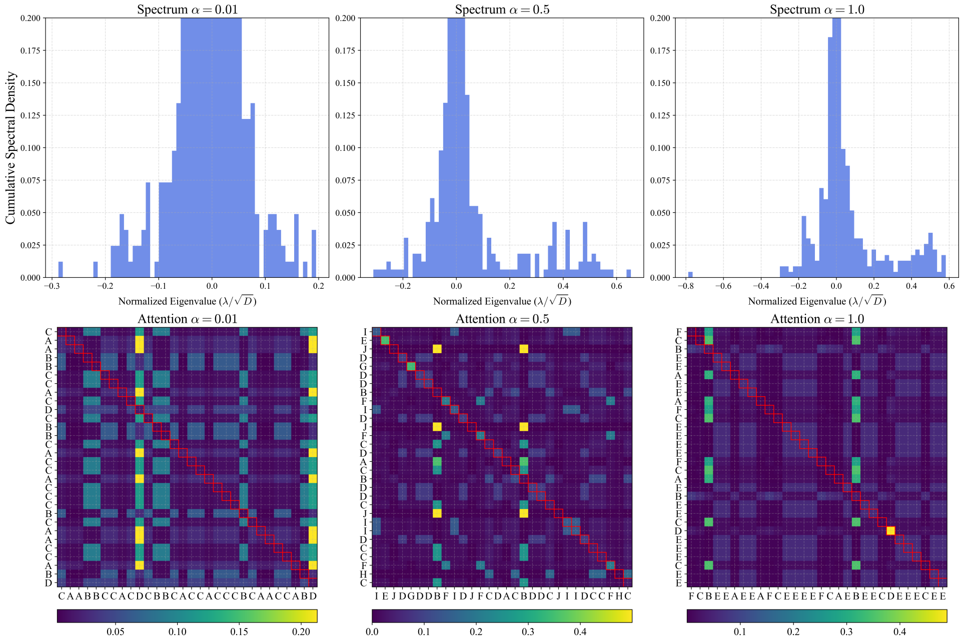


Figure 14: **Unfactorized Attention: Analysis of a Failure Mode. Top:** Eigenvalue spectra showing a massive peak at zero at $\alpha = 1.0$. The model has collapsed to a low-rank solution, but often the wrong one. **Bottom:** A representative attention matrix from a failed run. Instead of the semantic diagonal, we observe vertical striations. This indicates that the optimization got trapped in a "Saliency" minimum (attending to specific tokens globally) rather than finding the "Similarity" solution. This instability contrasts with the Tied model, which robustly finds the diagonal structure across all seeds. **Settings:** Student network with $d = 100$, $L = 15$, $T = 30$, optimized via Adam ($\eta = 0.01$).

Functional Analysis: The Robust Saliency Trap. Contrary to the Tied model which consistently learns the semantic solution, the Unfactorized model consistently settles into a suboptimal minimum characterized by vertical striations (Figure 14). Despite varying the initialization across 10 runs, the low variance in test accuracy (Figure 2) demonstrates that this 'Saliency' solution is a robust attractor of the optimization landscape for unstructured matrices, consistent with the saddle-to-saddle dynamics observed in [31]. The model reliably learns to exploit global token frequency (saliency) to achieve partial accuracy ($\approx 55\%$), but structurally fails to discover the pairwise counting mechanism.

Conclusion. Crucially, regardless of whether this collapse is driven by feature saliency or initialization artifacts, the outcome is the same: without the PSD constraint, the semantic solution is not the sole robust attractor of the dynamics.

K Detailed Analysis: Symmetric Unfactorized Attention

We analyze the Symmetric Unfactorized architecture ($S = S^\top$). Unlike the Tied model which enforces the Positive Semi-Definite (PSD) constraint by construction ($S = WW^\top$), this model only enforces symmetry.

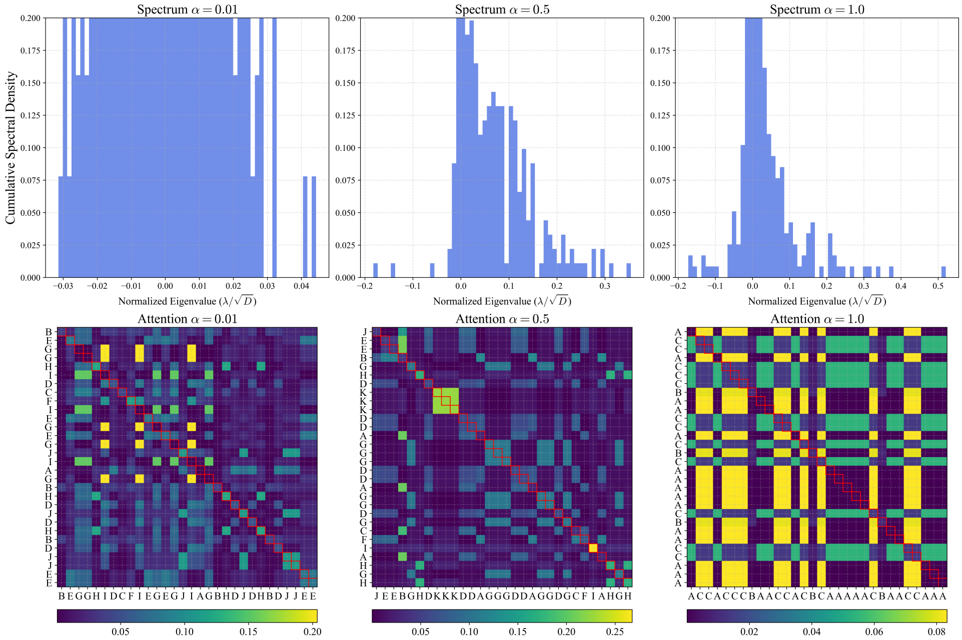


Figure 15: **Instability of Symmetric Attention (Representative Failure Case). Top:** Eigenvalue distribution showing persistent negative values. **Bottom:** A sample attention matrix exhibiting the "Saliency Trap" (vertical striations). Note that this is one outcome of a bimodal optimization landscape; while the model can theoretically learn the task, it lacks the robustness of the Tied architecture and frequently collapses into this non-semantic regime.

Optimization Landscape. While symmetry reduces the search space, Figure 2 shows that it does not guarantee convergence to the semantic solution. The model typically stagnates at an intermediate performance level, failing to eliminate negative eigenvalues. This confirms that without the explicit PSD constraint, the gradient dynamics are insufficient to escape the basin of attraction of the Saliency solution.

Figure 15 illustrates a representative example of this failure mode. The persistence of negative eigenvalues (top panel) and the vertical columnar structure (bottom panel) demonstrate that without the explicit PSD parameterization, the semantic solution is not a robust attractor of the gradient dynamics. The Tied model is therefore superior not because it has more capacity, but because it eliminates this parasitic failure mode.